Name
Date
Course

Absolute Value Equations and Inequalities (Homework)

Solve. **Answers**

① $8 = |x| + 7$ $\boxed{x = 1 \text{ or } -1}$

② $\dfrac{|y|}{2} = 10$ $\boxed{y = 20 \text{ or } -20}$

③ $-9 = -6|b| + 2$ $\boxed{b = \frac{11}{6} \text{ or } -\frac{11}{6}}$

④ $3 = 5 - \dfrac{|x|}{8}$ $\boxed{x = -16 \text{ or } 16}$

⑤ $|y - 7| = 5$ $\boxed{y = 12 \text{ or } 2}$

⑥ $-2 = \left|\dfrac{2b}{3}\right|$ $\boxed{\text{No Solution}}$

⑦ $|-5x| = 30$ $\boxed{x = -6 \text{ or } 6}$

⑧ $6 = 2|3y + 1| - 8$ $\boxed{y = 2 \text{ or } -\frac{8}{3}}$

⑨ $\dfrac{|-4 + 9b|}{6} - 3 = 7$ $\boxed{b = \frac{64}{9} \text{ or } -\frac{56}{9}}$

⑩ $-5 = 2 - 8\left|\dfrac{x}{10} - 1\right|$ $\boxed{x = \frac{75}{4} \text{ or } \frac{5}{4}}$

⑪ $\dfrac{\left|\frac{y}{-7} + 3\right|}{5} + 17 = 4$ $\boxed{\text{No Solution}}$

⑫ $-5|2x + 1| + 4 = 4$ $\boxed{x = -\frac{1}{2}}$

⑬ $2 = -7 + \dfrac{8|-6 - 5y|}{9}$ $\boxed{y = -\frac{129}{40} \text{ or } \frac{33}{40}}$

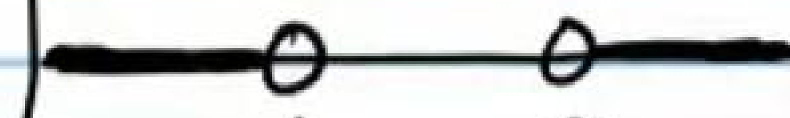
⑭ $3\left|5 - \dfrac{b}{2}\right| - 4 = 14$ $\boxed{b = -2 \text{ or } 22}$

⑮ $-8|2x| + 5 = 0$ $\boxed{x = \frac{5}{16} \text{ or } -\frac{5}{16}}$

⑯ $4|5 - 5y| + 3 = 6$ $\boxed{y = \frac{23}{20} \text{ or } \frac{17}{20}}$

Find the algebraic solutions and the graphical solutions of the following inequalities.

⑰ $5|2 + 4b| - 8 > 2$ $\boxed{b > 0 \text{ or } b < -1}$

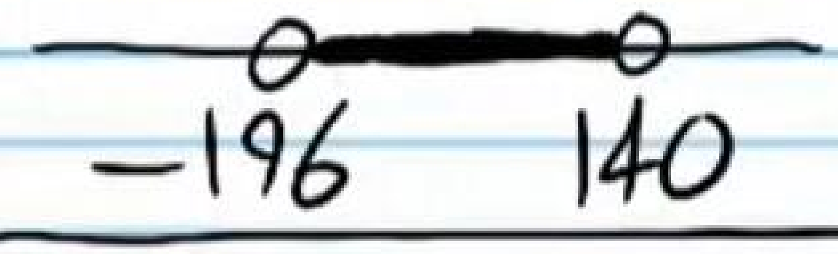


⑱ $2 \le -3\left|1 + \dfrac{y}{8}\right| + 5$ $\boxed{y \le 0 \text{ and } y \ge -16}$

⑲ $\left|\dfrac{x}{5} - 2\right| + 7 \le -7$ $\boxed{\text{No solutions}}$

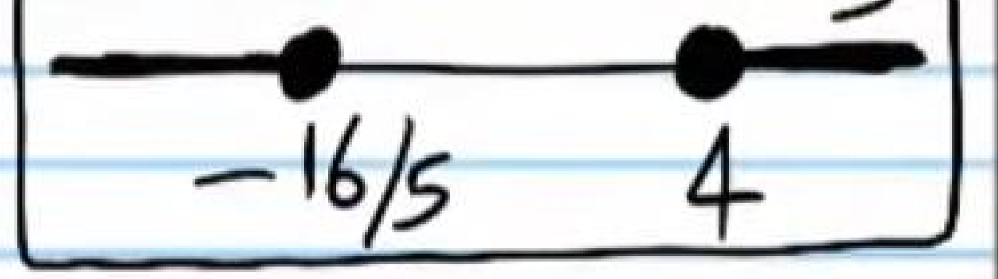
(20) $2 > -5 + \dfrac{\left|7 + \frac{b}{4}\right|}{6}$

$\boxed{b > -196 \text{ and } b < 140}$



(21) $-10 \geq 8 - |5x - 2|$

$\boxed{x \geq 4 \text{ or } x \leq -\frac{16}{5}}$

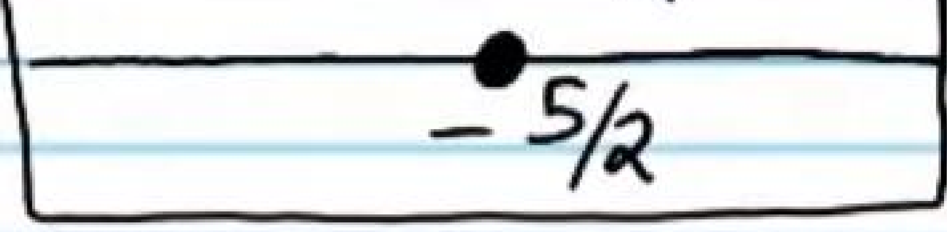


(22) $\dfrac{|y-2|}{4} - 7 > -9$

$\boxed{\text{All real numbers}}$

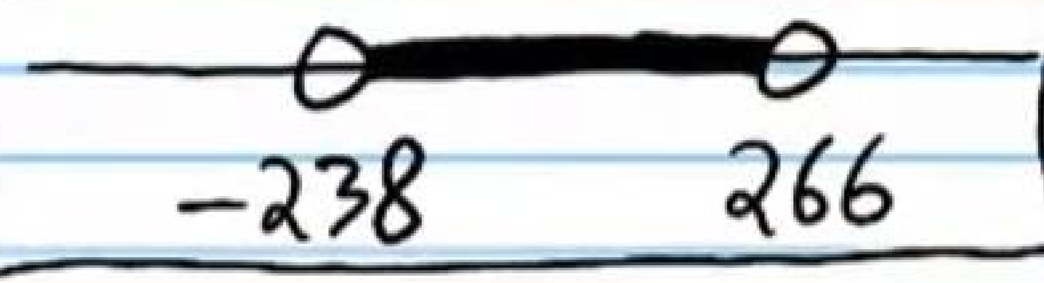
(23) $7 \leq 7 - 2|5 + 2b|$

$\boxed{b = -\frac{5}{2}}$



(24) $-2 < \dfrac{\left|\frac{x}{-7} + 2\right|}{-3} + 10$

$\boxed{x > -238 \text{ and } x < 266}$

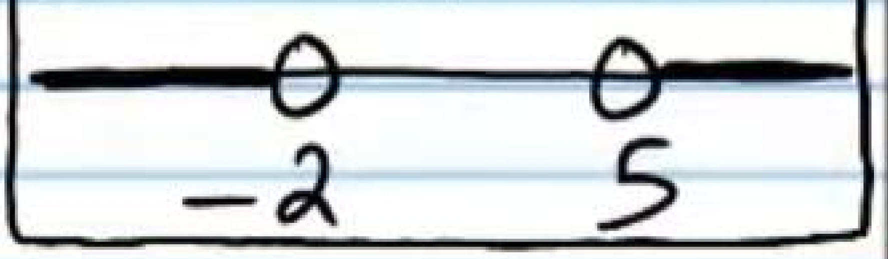


(25) $-3 > \left|\frac{y}{4} + 6\right| - 3$

$\boxed{\text{No Solutions}}$

(26) $\dfrac{-8|3 - 2b|}{7} + 5 < -3$

$\boxed{b > 5 \text{ or } b < -2}$

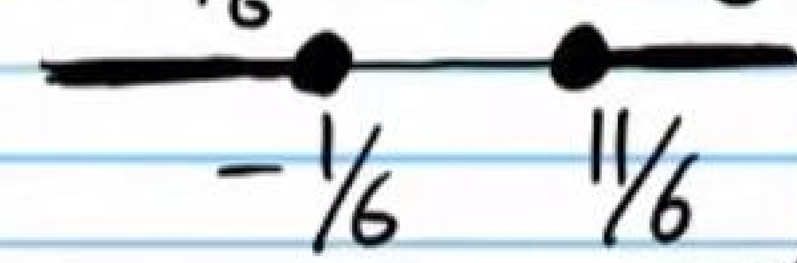


(27) $-8 \leq -8 + |x + 9|$

$\boxed{\text{All real numbers}}$

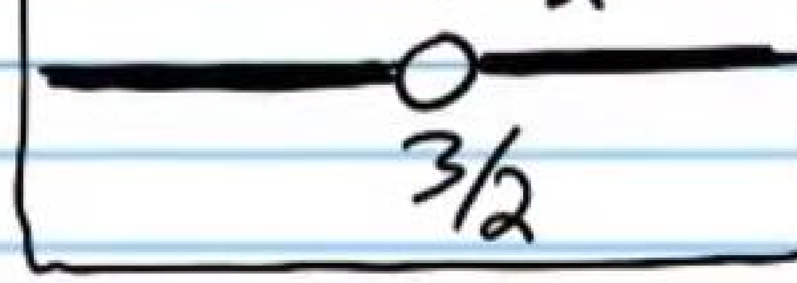
(28) $\dfrac{|6y - 5|}{-2} + 4 \leq 1$

$\boxed{y \geq \frac{11}{6} \text{ or } y \leq -\frac{1}{6}}$



(29) $0 > -\dfrac{|-2b + 3|}{7}$

$\boxed{b \neq \frac{3}{2}}$



(30) $6|-4x + 10| - 4 < 5$

$\boxed{x < \frac{23}{8} \text{ and } x > \frac{17}{8}}$